

Hai Duong

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SUMMARY

Applied Scientist Intern at AWS and PhD student in Computer Science at George Mason University specializing in automated reasoning, and scalable verification of deep neural networks. Lead developer of NeuralSAT, an open-source engine ranked 2nd at VNN-COMPs, with published papers at CVPR, NeurIPS, CAV, and FSE.

EDUCATION

- **PhD**, Computer Science, [George Mason University](#), Fairfax, VA 2022–present
- **MS**, Electrical Engineering, [Hanoi University of Science and Technology](#), Hanoi, VN 2021
- **BS**, Electrical Engineering, [Hanoi University of Science and Technology](#), Hanoi, VN 2019

EXPERIENCE

- **Applied Scientist Intern**, [Amazon Web Services](#), *Santa Clara, CA* Summer 2026
 - Developed SMT-based path enumeration and fuzzing test generation for compliance rules of AWS services in ControlsDSL, achieving 100% test coverage and exposing behavioral divergences between legacy and migrated rules.
 - Designed automated cross-compliance migration framework porting Java/Python to ControlsDSL and automatically verifies bidirectional behavioral equivalence. Migrated successfully 40% of staging rules to ControlsDSL for production.
- **Graduate Researcher**, [ROARS Lab](#), [George Mason University](#), *Fairfax, VA* 2022–present
 - Architected [NeuralSAT](#) (Python, PyTorch, CUDA), an open-source GPU-accelerated DNN verification engine; ranked 4th (2023, 2026) and 2nd (2024, 2025) at VNN-COMP.
 - Developed [VeriStable](#) (FSE'24), a DPLL-based verifier exploiting stable neurons; outperforms α - β -CROWN and NeuralSAT and solved 48 of 51 hard GDVB problems vs. 4 for the baseline (12 $\times$ more).
 - Extended robustness verification to new perturbation classes with [VeriS](#) (FSE'26) and [VeriDou](#) (CVPR'26 Highlight): VeriS verifies 78% of 5,508 structural-robustness problems prior tools could not, while VeriDou exposes adversarial examples over 14,340 instances on networks prior methods claimed robust.
 - Designed [APTP](#) (NeurIPS'25), a verifier-independent proof format whose checker outperforms prior work across 16 networks and 400 problems.
 - Created Cabl, a constrained abstract domain using a Lagrangian primal–dual method; over 8,640 instances, verifies up to 19% more properties than the best prior abstraction and 31% more on constrained properties at 5.7 $\times$ the speed.
 - Built VerCa, an autoencoder robustness verifier solving 87% of 1,500 latent-space properties vs. 66–67% for state-of-the-art verifiers, with zero invalid counterexamples.
- **Research Intern**, [BachLe's Lab](#), University of Melbourne, *Remote* 2021–2022
 - Built FFL, a graph-neural fault-localization model for student code that improved top-1 accuracy by up to 197% over prior baselines and localized bugs within the top-10 lines for 83–85% of programs (ICSME 2022).

SKILLS

- **Programming Languages:** Python, C/C++, MATLAB, SMTLib
- **ML & Systems:** PyTorch, CUDA/GPU acceleration, NumPy
- **Methods:** SAT/SMT solving, abstract interpretation, branch-and-bound, LP/MILP, reinforcement learning

SELECTED PUBLICATIONS (FULL LIST)

- P1 Duong et al., “*Verifying Neural Networks with Reinforcement Learning*”, NeurIPS 2026.
P2 Duong et al., “*Verifying Neural Network Robustness with Dual Perturbations*”, CVPR 2026.
P3 Duong et al., “*Verifying Structural Robustness of Deep Neural Network*”, FSE 2026.
P4 Duong et al., “*Compositional Neural Network Verification via Assume-Guarantee Reasoning*”, NeurIPS 2025.
P5 Duong et al., “*Generating and Checking DNN Verification Proofs*”, NeurIPS 2025.
P6 Duong et al., “*NeuralSAT: A High-Performance Verification Tool for Deep Neural Networks*”, CAV 2025.
P7 Duong et al., “*Harnessing Neuron Stability to Improve DNN Verification*”, FSE 2024.
P8 Xu et al., “*Training for Verification: Increasing Neuron Stability to Scale DNN Verification*”, TACAS 2024.

AWARDS

- Tinker Research Grants (\$5,000 API Credits), Thinking Machines 2026
- Outstanding PhD Student Award, GMU 2026
- Highlight Paper Award [P2], CVPR 2026
- Spotlight Paper Award [P4], NeurIPS 2025